REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 10 Advanced | Module 16 Lesson 3 | Refraction of Light

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: n = c/v; n1 sin i = n2 sin r; sin c = n2/n1 for n1 > n2. Angles are measured from the normal.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 19–25.

P01. When light enters glass from air, its frequency:

A. Always increases B. Always decreases
C. Becomes zero D. Remains unchanged

Answer

D. Remains unchanged.

P02. Total internal reflection requires light to travel:

A. From higher index to lower index above the critical angle B. From lower index to higher index at any angle
C. Along the normal only D. Through a vacuum only

Answer

A. From higher index to lower index above the critical angle.

P03. At the critical angle, the refracted ray is:

A. Along the normal B. At 90° to the normal
C. At the same angle as the incident ray D. Absent because the frequency is zero

Answer

B. At 90° to the normal.

P04. Dispersion by a prism occurs because refractive index depends on:

A. Beam area only B. The observer's height
C. Wavelength D. The prism's mass only

Answer

C. Wavelength.

P05. A ray entering a higher-index medium obliquely usually bends:

A. Away from the normal B. Through 90° always
C. Along the boundary always D. Toward the normal

Answer

D. Toward the normal.

P06. Light travels at 2.14×10^8 m/s in a transparent medium. Calculate its refractive index using $c = 3.00 \times 10^8$ m/s.

A. 2.804 B. 0.7009
C. 1.402 D. -1.402

Answer

C. 1.402. $n = c/v = (3.00 \times 10^8)/(2.14 \times 10^8) = 1.402$.

P07. Light enters glass (n = 1.50) from air (n = 1.00) at 25° to the normal. Find the refraction angle.

A. 32.73 ° B. 8.182 °
C. -16.36 ° D. 16.36 °

Answer

D. 16.36 °. $n1 \sin i = n2 \sin r$; $r = \sin^{-1}(\sin 25°/1.50) = 16.36°$.

P08. Light passes from a medium of index 1.50 toward air (n = 1.00). Calculate its critical angle.

A. 41.81 ° B. 83.62 °
C. 20.91 ° D. -41.81 °

Answer

A. 41.81 °. $\sin c = n2/n1 = 1.00/1.50$; $c = 41.81°$.

P09. A medium has refractive index 1.40. Calculate light speed in it. Use $c = 3.00 \times 10^8$ m/s.

A. 4.29×10^8 m/s B. 2.14×10^8 m/s
C. 1.07×10^8 m/s D. -2.14×10^8 m/s

Answer

B. 2.14×10^8 m/s. $v = c/n = (3.00 \times 10^8)/1.40 = 2.14 \times 10^8$ m/s.

P10. Light of wavelength 550 nm in air enters glass of refractive index 1.50. Find its wavelength in glass.

A. 733.3 nm B. 183.3 nm
C. 366.7 nm D. -366.7 nm

Answer

C. 366.7 nm. Frequency is unchanged; $\lambda_{medium} = \lambda_{air}/n = 550/1.50 = 366.7$ nm.

P11. Define refraction.

Answer

A change in wave propagation direction at a boundary due to a change in wave speed, except when incidence is normal.

P12. Define refractive index.

Answer

The ratio of light speed in vacuum to its speed in a medium: $n = c/v$.

P13. Define critical angle.

Answer

The incidence angle in the higher-index medium for which the refracted ray is at 90° to the normal.

P14. State both conditions for total internal reflection.

Answer

Light must travel toward a lower-index medium, and its incidence angle must exceed the critical angle.

P15. Explain the main cause of a mirage near a hot road.

Answer

A temperature-dependent refractive-index gradient bends light through layers of air, making light from the sky appear to come from the road.

P16. Light travels at 2×10^8 m/s in a transparent medium. Calculate its refractive index using $c = 3.00 \times 10^8$ m/s.

Answer

$n = c/v = (3.00 \times 10^8)/(2 \times 10^8) = 1.5$.

P17. Light travels at 1.88×10^8 m/s in a transparent medium. Calculate its refractive index using $c = 3.00 \times 10^8$ m/s.

Answer

$n = c/v = (3.00 \times 10^8)/(1.88 \times 10^8) = 1.596$.

P18. Light enters glass (n = 1.50) from air (n = 1.00) at 30° to the normal. Find the refraction angle.

Answer

$n_1 \sin i = n_2 \sin r$; $r = \sin^{-1}(\sin 30°/1.50) = 19.47°$.

P19. Light enters glass (n = 1.50) from air (n = 1.00) at 35° to the normal. Find the refraction angle.

Answer

$n_1 \sin i = n_2 \sin r$; $r = \sin^{-1}(\sin 35°/1.50) = 22.48°$.

P20. Light passes from a medium of index 1.60 toward air (n = 1.00). Calculate its critical angle.

Answer

$\sin c = n_2/n_1 = 1.00/1.60$; $c = 38.68°$.

P21. Light passes from a medium of index 1.70 toward air (n = 1.00). Calculate its critical angle.

Answer

$\sin c = n_2/n_1 = 1.00/1.70$; $c = 36.03°$.

P22. A medium has refractive index 1.50. Calculate light speed in it. Use $c = 3.00 \times 10^8$ m/s.

Answer

$v = c/n = (3.00 \times 10^8)/1.50 = 2 \times 10^8$ m/s.

P23. A medium has refractive index 1.60. Calculate light speed in it. Use $c = 3.00 \times 10^8$ m/s.

Answer

$v = c/n = (3.00 \times 10^8)/1.60 = 1.88 \times 10^8$ m/s.

P24. Light of wavelength 600 nm in air enters glass of refractive index 1.50. Find its wavelength in glass.

Answer

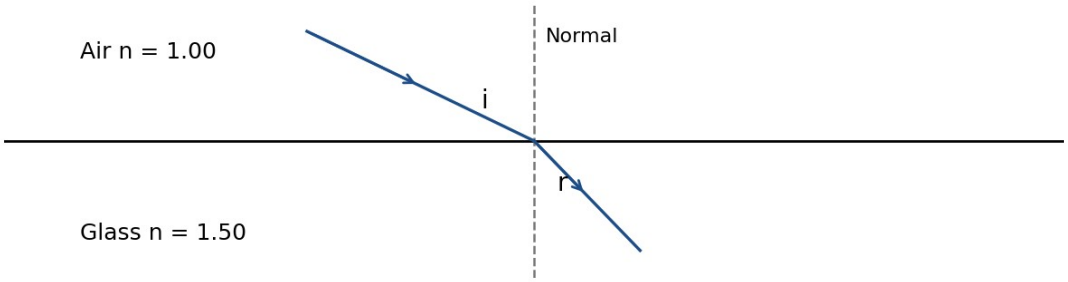
Frequency is unchanged; $\lambda_{medium} = \lambda_{air}/n = 600/1.50 = 400$ nm.

P25. Light of wavelength 650 nm in air enters glass of refractive index 1.50. Find its wavelength in glass.

Answer

Frequency is unchanged; $\lambda_{medium} = \lambda_{air}/n = 650/1.50 = 433.3$ nm.

P26. Draw a ray entering glass obliquely from air. Label incident ray, refracted ray, boundary, normal, i and r.



Answer

Draw the refracted ray closer to the normal because glass has a higher index. Label i in air and r in glass, both measured from the normal.

P27. For glass of index 1.50 into air, compare incidence angles 40° and 50°. Which gives total internal reflection?

Answer

The critical angle is $\sin^{-1}(1/1.50) \approx 41.8°$. At 40° some light refracts into air; at 50° total internal reflection occurs.

P28. Can light change speed at a boundary without changing direction? Explain.

Answer

Yes. At normal incidence it travels along the normal without bending, although its speed and wavelength change; its frequency remains unchanged.

P29. Why does white light split into colours through a prism?

Answer

Different wavelengths experience different refractive indices and therefore different refraction angles. The component colours spread out.

P30. Design a method to estimate glass refractive index using a ray box.

Answer

Trace the boundary and normal, measure i and r for several nonzero angles, calculate $n \approx \sin i / \sin r$ for air-to-glass rays and compare repeated values.